

Generative AI and Labor Demand: Evidence from Online Job Markets

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- The empirical record on AI and employment is surprisingly mixed:
 - Firm surveys: widespread adoption, modest employment effects (Yotzov et al., 2026)
 - Administrative data: precise null on earnings and hours (Humlum and Vestergaard, 2025)
 - Freelance platforms: meaningful contractions for writing- and coding-intensive tasks (Demirci et al., 2025; Hui et al., 2023)
- Much of the difficulty stems from the granularity and timeliness of available data, especially outside the US.
- This paper: high-frequency online job postings across the EU (CEDEFOP Skills OVATE), five task-based AI exposure measures, ChatGPT release as the event date.

Are occupations with higher exposure to AI experiencing worse labor demand outcomes relative to occupations with lower exposure?

Relative to the existing empirical literature:

- European evidence to complement the largely US-focused work
- Within-occupation skill-level analysis, not only across occupations
- Heterogeneity by experience level
- Out-of-sample validation on Australian data

Key Findings

- Occupations at the highest levels of AI exposure see a **19–29% relative decrease** in EU job postings since the release of ChatGPT, robust across five exposure measures.
- Partially replicated out-of-sample on Australian data (the two most recent measures).
- Within occupations, skills most similar to AI capabilities are mentioned **12% less frequently**.
- **No significant impact** on industry-level labor or capital productivity, consistent with “so-so automation”.
- Inverted U-shape in experience: the decline concentrates in entry-level postings and those requiring 10+ years of experience.

Task-Based Framework

- Task-based model of automation (Zeira, 1998; Acemoglu et al., 2022): sector output is an aggregate of task outputs, each produced by labor or AI.
- Sector exposure = share of the labor force employed in tasks about to be automated:

$$d \ln l_s = (-1 + (\epsilon_s \rho_s - 1) \pi_s) \times exposureAI_s$$

- The sign is theoretically ambiguous:
 - Displacement dominates $\Rightarrow$ employment falls with exposure
 - Large enough productivity gains (elastic demand, high pass-through) $\Rightarrow$ employment can expand
- Which effect dominates is an empirical question for any given technology.

Five Measures of Occupational AI Exposure

Measure	Source data	Vintage
Felten et al. (2018)	Crowd-sourced ability–application links	pre-LLM
Webb (2022)	AI patents matched to O*NET tasks	pre-LLM
Eloundou et al. (2023)	GPT-4 rates its own automation potential	post-LLM
Demirev (2026)	AI product press releases vs. ESCO skills	post-LLM
Handa et al. (2025)	Observed Claude.ai usage by O*NET task	post-LLM

- All rescaled to $[0, 1]$; results reported separately for each measure.
- Demirev (2026) and Handa et al. (2025) additionally split exposure into “automation” vs. “augmentation”.
- Demirev (2026) also provides skill-level exposure, used for the within-occupation analysis.

- **Skills OVATE (CEDEFOP/Eurostat):** 100M+ online job adverts across the EU, Q4 2021–Q2 2025; occupations (ESCO level 3) and skill mentions by country.
- **EURES:** vacancies by required experience level; only available from Q3 2024, so used descriptively.
- **Australian Internet Vacancy Index:** ANZSCO 4-digit vacancies by state, 2016–2026; a full decade of pre-trends.
- **Eurostat national accounts (nama10):** labor and capital productivity by NACE industry, 2021–2024.
- Caveat: online job openings are only a proxy of real labor demand (“ghost listings”, uneven coverage across occupations).

The Overall Market Contracted Sharply

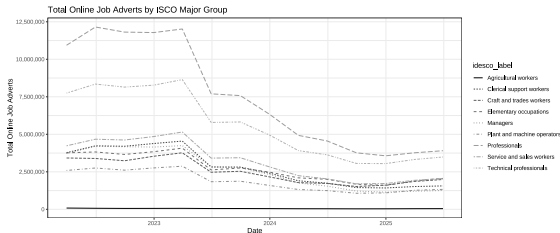


Figure: Total online job adverts by ESCO level-1 occupation (rolling 12-month totals). Source: Cedefop (2021)

From 40.3M adverts (early 2022) to 17.7M (mid 2025), a 50%+ decline, steeper than any comparable benchmark. All specifications compare occupations *within* each period.

Long-Difference Specification

Difference-in-differences with continuous treatment; event date is the release of ChatGPT (November 2022):

$$\Delta \log \bar{y}_{c,i} = \delta_0 + \xi_c + \delta X_i^{AI} + \varepsilon_{c,i}$$

- $\Delta \log \bar{y}_{c,i}$: change in log average job postings per quarter, pre vs. post event, for occupation i in country c
- X_i^{AI} : one of the five exposure measures, scaled to $[0, 1]$
- ξ_c : country fixed effects; standard errors clustered by country
- Event-study version with country-occupation fixed effects tests for pre-trends
- Analogous specifications for skills (occupation FE added), experience levels, and industry productivity

Main Result: AI Exposure and Job Postings

Table: Impact of AI Exposure on Online Job Ads ($\Delta \log(\text{OJA})$)

	(1)	(2)	(3)	(4)	(5)
Demirev Exposure	-0.211*** (0.055)				
Felten AI Exposure		-0.212** (0.058)			
Webb AI Exposure			-0.205*** (0.043)		
Eloundou Exposure				-0.289*** (0.066)	
Anthropic Usage					-0.339*** (0.054)
Observations	2,788	2,785	2,776	2,785	2,785
R ² (within)	0.013	0.026	0.016	0.034	0.017
Country FE	Yes	Yes	Yes	Yes	Yes

Standard errors clustered at country level. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Moving from lowest to highest exposure: **19–29% relative decrease** in postings.
Significant for all five measures.

Visual Illustration: Partial Correlation

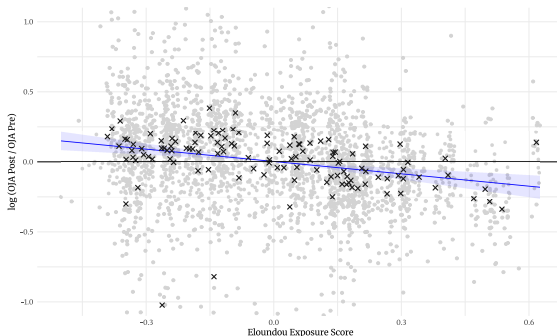


Figure: Partial correlation of AI exposure and $\Delta \log(\text{OJA})$, residualized on country FE (Eloundou measure). Dots: country-occupation pairs; crosses: occupation averages.

Within-country R^2 is small (0.013–0.034): AI exposure explains little of the overall contraction, but predicts *where* it concentrates.

Event Study: The Effect Grows Over Time

Event Study Results Across Different AI Exposure Measures

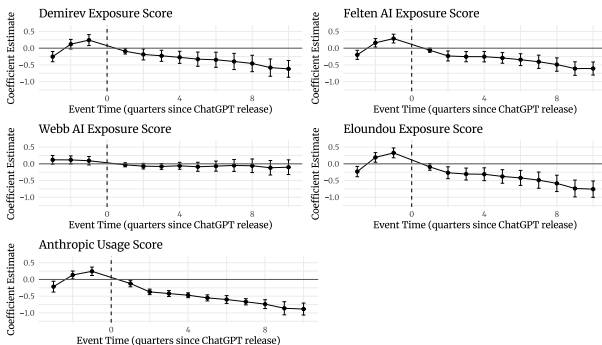


Figure: Event-study coefficients around the ChatGPT release, all five measures.

Significant negative coefficients from 2 quarters post-event, growing to 46–59% by mid-2025 (Webb is the exception). Some significant pre-event coefficients; addressed with the longer Australian pre-period.

Automation vs. Augmentation

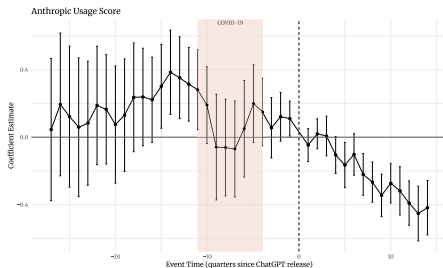
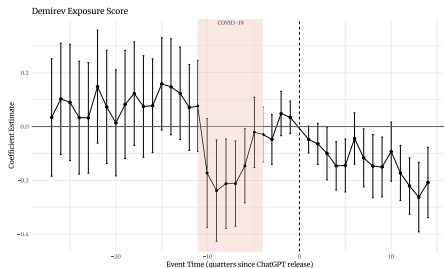
Table: Impact on OJA by Intent ($\Delta \log(\text{OJA})$)

	Handa	Demirev
Automation Exposure	-0.236** (0.064)	0.064 (0.050)
Augmentation Exposure	0.137* (0.059)	-0.277*** (0.069)
Observations	2,785	2,788
Country FE	Yes	Yes

Clustered (by country) SEs. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

- Handa et al. (2025): demand falls for automatable jobs, rises where AI augments workers.
- Demirev (2026) does not separate as cleanly; its intent breakdown should be treated as suggestive.

Out-of-Sample Validation: Australia



- The two most recent measures replicate: Anthropic Usage -0.357 (0.036), Demirev -0.152 (0.025), i.e. 14–30% relative reduction.
- Felten is significant but *positive*; Webb and Eloundou insignificant.
- A decade of pre-trends, reaching through COVID (shaded): no evidence the post-ChatGPT decline is reversion-to-the-mean.

Skill-Level Demand Within Occupations

Dep. var.:	$\Delta \log(\text{Skill Mentions})$
AI Product Exposure	-0.246* (0.113)
Observations	75,669
Occupation FE	Yes
Country FE	Yes

SEs clustered by country and occupation.

- Most exposed skill in the sample: roughly **12% fewer mentions** post-GPT, net of occupational composition (occupation FE).
- Suggests changes in task composition within occupations.
- Caveat: significant pre-event coefficients in the event study, so the causal interpretation is weaker here.

Experience Heterogeneity: An Inverted U-Shape

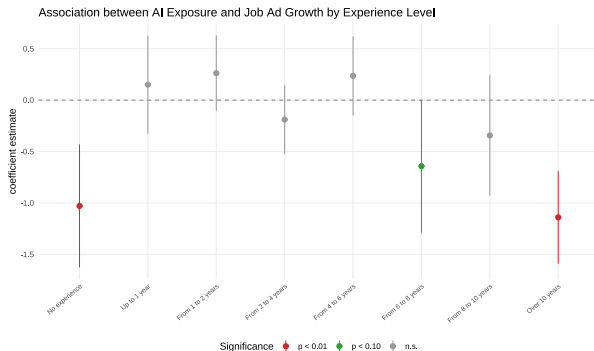


Figure: Exposure gradients by experience bin, Handa et al. (2025) measure (EURES data, 2024Q3–2025Q2, descriptive).

Largest declines at both tails: entry-level *and* 10+ years of experience (36–67% fewer vacancies at highest exposure). Entry-level echoes Brynjolfsson et al. (2025); the senior tail is new.

Productivity: A Null Result

- Industry exposure scores built from the occupational mix within each industry (ESCO employment weights).
- TWFE and long-difference models of labor and capital productivity (Eurostat, 2021–2024).
- **None of the estimated coefficients is significant** for any of the five measures.
- The null is informative: displacement appears to operate without a measurable productivity offset at sector-level granularity.
- Caveats: annual frequency, series ends six months before the OJA data, only 17 industries.

- **Leave-one-out:** dropping any single occupation (level 2 or 3) or country never flips the sign or loses significance, for any measure.
- **TWFE in levels:** all five interaction coefficients negative and significant (-0.168 to -0.483).
- **Deciles:** broadly monotone; declines concentrated in the top exposure deciles.
- **Teleworkability confound:** the two measures *least* correlated with Dingel and Neiman (2020) teleworkability (Demirev, Handa) are exactly the ones that replicate on Australian data, arguing against a remote-work reversion story.

Conclusion: “So-So Automation”?

- Highly exposed occupations show a **19–29% relative decline** in job postings post-ChatGPT, growing over time, robust across five exposure measures, and partially replicated in Australia.
- Within occupations, AI-similar skills are mentioned up to 12% less.
- No measurable industry-level productivity effect.
- Together, consistent with Acemoglu and Restrepo (2019)’s “so-so automation”: AI does enough to displace parts of labor, but the efficiency gains are insufficient to offset the disemployment effects.
- Initial effects only; monitoring labor market outcomes for exposed occupations should remain a priority.

Caveats and Limitations

- AI exposure explains little of the *overall* market contraction; it predicts its distribution across occupations.
- The post-ChatGPT period coincides with monetary tightening and the unwinding of pandemic-era policies; interest-rate-sensitive sectors (e.g. software) rank high on every exposure measure.
- Some significant pre-event coefficients in the EU event studies; alleviated but not eliminated by the Australian pre-period and the teleworkability analysis.
- Online job adverts are a partial view of the labor market: postings may decline even if employment stays fixed.

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